An Introduction to ODEs

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Examples of ODEs

Definition (Differential Equation)

A *differential equation* is an equation involving one or more derivatives of an unknown function.

$y_1 + y_2$

Example (Example of ODEs)

The following are all examples of differential equations:

y_1, y_2

1. $\frac{\mathrm{d}y}{\mathrm{d}x} + y = x^2.$
2. $\sin\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) + \tan^{-1} y = 1.$ } ODE
3. $\phi_{xx} + \phi_{yy} - \phi_x = \mathrm{e}^x + x \sin y.$ } PDE

$\boxed{y = y(x)}$

$\frac{dy_i}{dx} + y_i = x^2$, $i = 1,2$

$\phi = \phi(x, y)$

Definition (Order of a DE)

The order of the highest derivative occurring in a differential equation is called the **order** of the differential equation.

order $\boxed{3}$ — $y''' + y = \sin(xy)$ / $(y''')^2 + y = \sin x$

linear eqⁿ

$$\frac{dy_1}{dx} + y_1 = x^2$$

$$\frac{dy_2}{dx} + y_2 = x^2$$

Is $y_1 + y_2$ a solⁿ of $y'' + y = x^2$?

Ans. No

$$\frac{dy_1}{dx} + y_1 = 0$$

$$\frac{dy_2}{dx} + y_2 = 0$$

Is $y_1 + y_2$ a solⁿ of $y'' + y = 0$

$\cancel{a_n(x) = 1}$

Equⁿ #1	Equⁿ #2
$a_1 = 1$ and $a_0 = 1$	$a_1 = 1$, $a_0 = 1$
$F(x) = x^2$	$F(x) = 0$

Linear ODEs

Definition (Linear Differential Equation)

A differential equation that can be written in the form

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = F(x), \tag{1}$$

where $a_0, a_1, \cdots, a_n$ and F are functions in x only, is called a linear differential equation of order n (such that $\boxed{a_0 \neq 0}$). Such a differential equation is linear in $y, y', y'', \cdots, y^{(n)}$.

Nonlinear Differential Equation

A differential equation that does not satisfy the above definition is called a nonlinear differential equation.

Nonlinear ODE:

$(y''')^2 + y = \sin xy$

$y'' + e^y = \sin xy$

$\sin(y'') + y = 0$

Example of Linear and Nonlinear ODEs

Example

The equations

$F(x,y) = \frac{2y}{(x^2+1)}$

$$y'' + x^2y' + (\sin x)y = e^x \text{ and } xy''' + 4x^2y' = \frac{2}{1+x^2}y$$

are linear ODEs of order 2 and 3, respectively. Whereas

$$y'' + x\sin(y') - xy = y^2 \text{ and } xy''' - x^2y + y^2 = 0$$

are nonlinear ODEs of order 2 and 3, respectively.

n.l. ODE.

2nd order nonlinear ODE.

non-linear

nonlin. 3rd order ODE.

$y'' + y = 0$; $\boxed{y \equiv y(x)}$

$$\boxed{y'' + y = F(x)}$$ ← inhomogeneous linear eqn

$y = y(x)$

$$f(x) = \begin{cases} x^2 \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$f'(x) = 2x \sin\frac{1}{x}$

$- \cos\frac{1}{x}$

$C^n(I)$ $+$ cont

Definition (Solution to an ODE)

A function $y = f(x)$ that is (at least) n-times differential on an interval I is called a solution to the differential equation (1) on I if the substitution $y = f(x), y' = f'(x), \cdots, y^{(n)} = f^{(n)}(x)$ reduces to the differential equation (1) to an identity valid for all $x \in I$. In this case we say that $y = f(x)$ satisfies the ODE.

$$y'' + y = F(x)$$

$y \in$ twice diff^ble $\Rightarrow y'' \notin$ cont^s

Examples of Solution to an ODE

Example

Verify that for all constants c_1 and c_2 $y(x) = c_1 \sin x + \cancel{c_2 \sin x}$ is a solution to the linear differential equation $y'' + y = 0$ for x in the interval $(-\infty, \infty)$.

$y(x) = c_1 \sin x$

$c_1 = 1, 2, -3, e^{-1}.$

Example

Example

Verify that the relation $x^2 + y^2 - 4 = 0$ defines an implicit solution to the nonlinear differential equation

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.}$$

General Solution of an ODE

Definition

A solution to an n-th order differential equation on an interval I is called the general solution on I if it satisfies the following conditions:

- The solution contains n constants $c_1, c_2, \cdots, c_n$.
- All solutions to the differential equation can be obtained by assigning appropriate values to the constants.

Remark

Not all differential equations have a general solution. For example, consider

$y-1=0$ in $\mathbb{R}$

$y \equiv 1$ in $\mathbb{R}$

in $\mathbb{R}$

$$(y')^2 + (y-1)^2 = 0.$$

$\Rightarrow y'=0,\ (y-1)=0$ in $\mathbb{R}$.

The only solution to this differential equation is $y(x) = 1$, and hence the differential equation does not have a solution containing an arbitrary constant.

Examples: General Solution

Example (Simplest Case)

Consider the ODE $\frac{\mathrm{d}y}{\mathrm{d}x} = f(x)$. Then, $y = f(x) + C$ is the general solution.

Example

Consider the ODE $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3x}{1+x^2}$. Then

$$y(x) = \int \frac{3x}{1+x^2} x = \frac{3}{2} \ln(1+x^2) + C.$$

–A general solution since C is not specified.

Population Growth Model

Example

Suppose a population grows at a rate which proportional to the size of the population. Let $P(t)$ denotes the population of some species at time t. Then we can model the population as in the following ODE:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = kP(t), \; k > 0.$$

Then $P(t) = C\mathrm{e}^{kt}$.

Radio Active Decay

Example

Let $Q(t)$=Amount of radioactive substance at time t. Then one of the simplest model to demonstrate the radioactive decay is considered as

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -kQ(t).$$

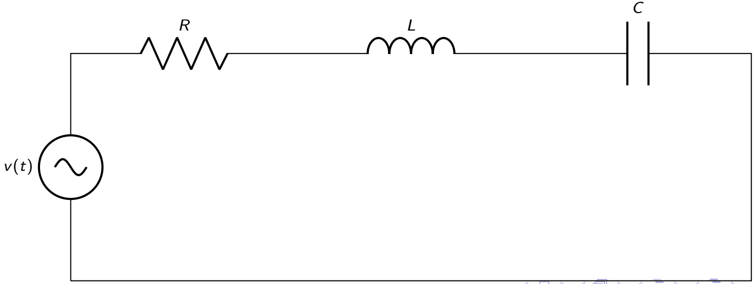
Remark

It is well known that most radioactive materials decay at a rate proportional to the amount present.

What is a First-Order Circuit?

- A **first-order circuit** contains:
  - One energy storage element:
    - Capacitor C, or
    - Inductor L
  - Any number of resistors and sources

- Because there is only one storage element, the governing equation contains only **one derivative**.



RL Circuit: Governing Equation

- Using Kirchhoff's Voltage Law (where $v(t)$ is the voltage source):

$$v(t) = v_R + v_L.$$

- We know (for the current $i(t)$):

$$v_R = iR, \qquad v_L = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

- Thus:

$$\boxed{v(t) = L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri}$$

- Rewriting:

$$\boxed{\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{R}{L}i = \frac{1}{L}v(t).}$$

Thank You

Questions?